Multi-Corner Static Timing Analysis

VSDBabySoC · Sky130 HD · 16 PVT corners

Author: MD Soahil Ahmed

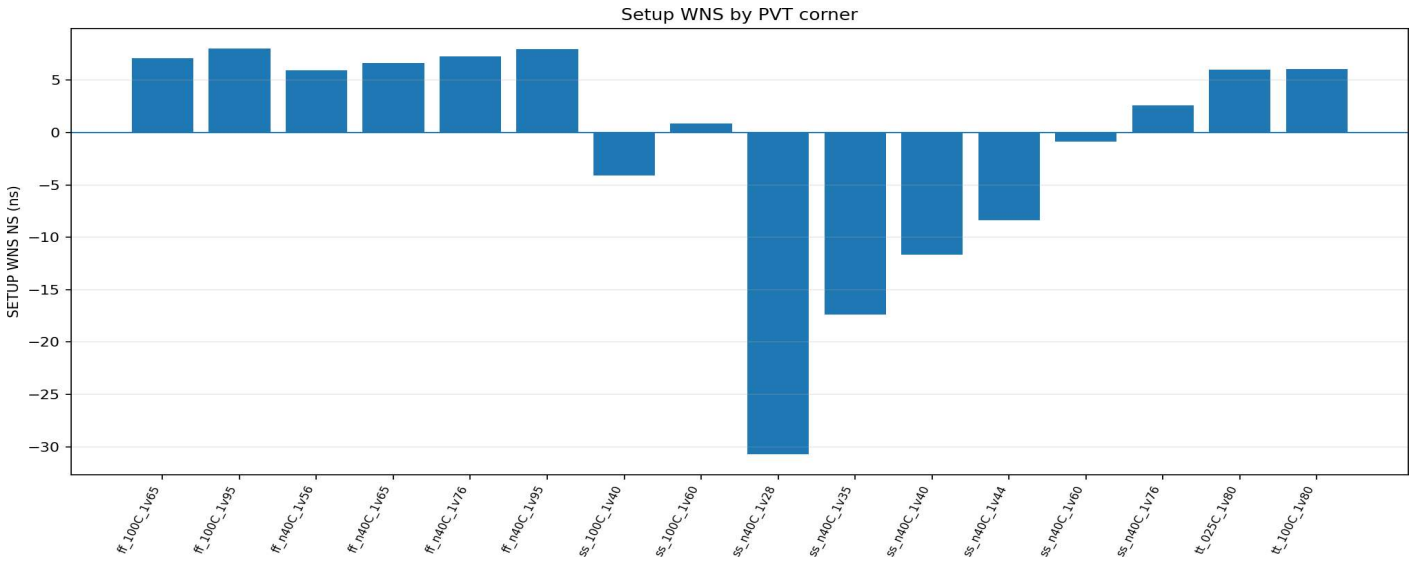
This report summarizes the available corner metrics and generated charts. It is not a substitute for original detailed timing path reports.

Metric	Result
Corners summarized	16
Corners with negative setup WNS	6
Corners with negative hold WNS	0
Worst setup WNS	−30.7252 ns · ss_n40C_1v28
Best setup WNS	+8.0138 ns · ff_100C_1v95
Lowest hold WNS	+0.1916 ns · ff_n40C_1v95

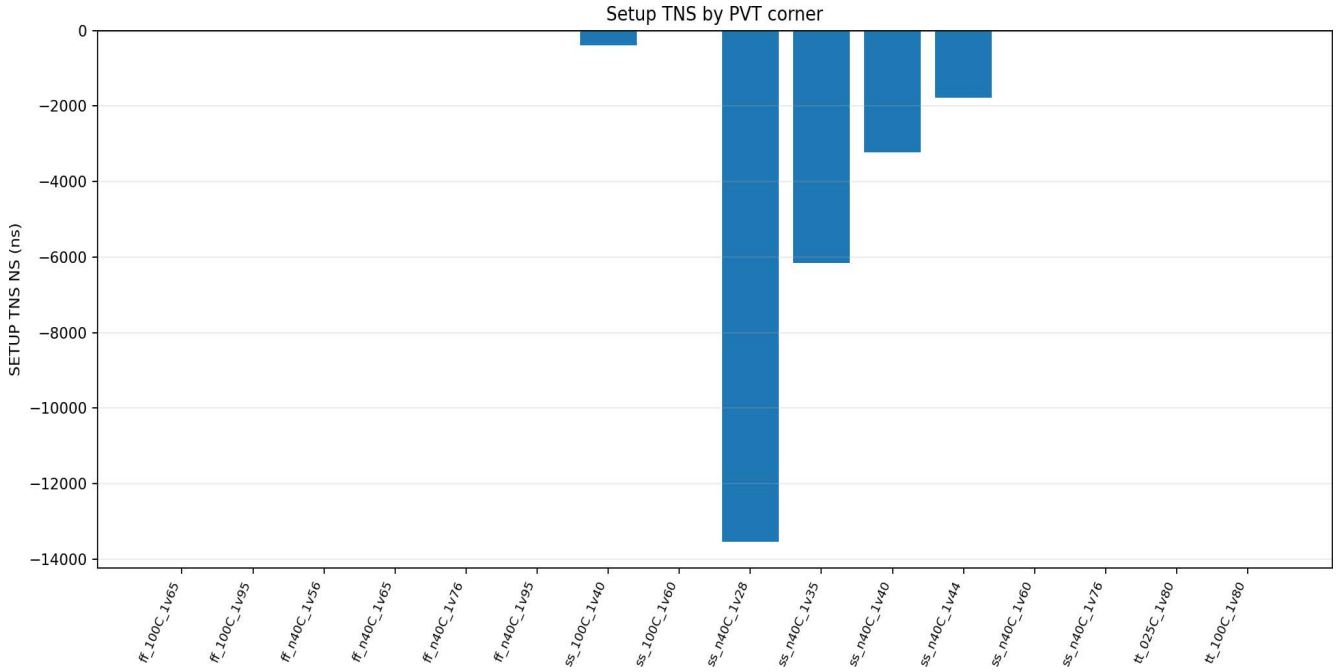
Corner metrics

Corner	Setup WNS	Setup TNS	Hold WNS	Hold TNS
ff_100C_1v65	7.1134	0.0000	0.2508	0.0000
ff_100C_1v95	8.0138	0.0000	0.1985	0.0000
ff_n40C_1v56	5.9507	0.0000	0.2916	0.0000
ff_n40C_1v65	6.6615	0.0000	0.2570	0.0000
ff_n40C_1v76	7.2816	0.0000	0.2274	0.0000
ff_n40C_1v95	7.9913	0.0000	0.1916	0.0000
ss_100C_1v40	-4.1254	-398.1285	0.9008	0.0000
ss_100C_1v60	0.8405	0.0000	0.6404	0.0000
ss_n40C_1v28	-30.7252	-13550.8096	1.7552	0.0000
ss_n40C_1v35	-17.3561	-6161.1196	1.3135	0.0000
ss_n40C_1v40	-11.6665	-3219.5767	1.0999	0.0000
ss_n40C_1v44	-8.3726	-1788.6748	0.9676	0.0000
Corner	**Setup WNS**	**Setup TNS**	**Hold WNS**	**Hold TNS**
ss_n40C_1v60	-0.8504	-17.4777	0.6539	0.0000
ss_n40C_1v76	2.6041	0.0000	0.4871	0.0000
tt_025C_1v80	5.9879	0.0000	0.3111	0.0000
tt_100C_1v80	6.0693	0.0000	0.3156	0.0000

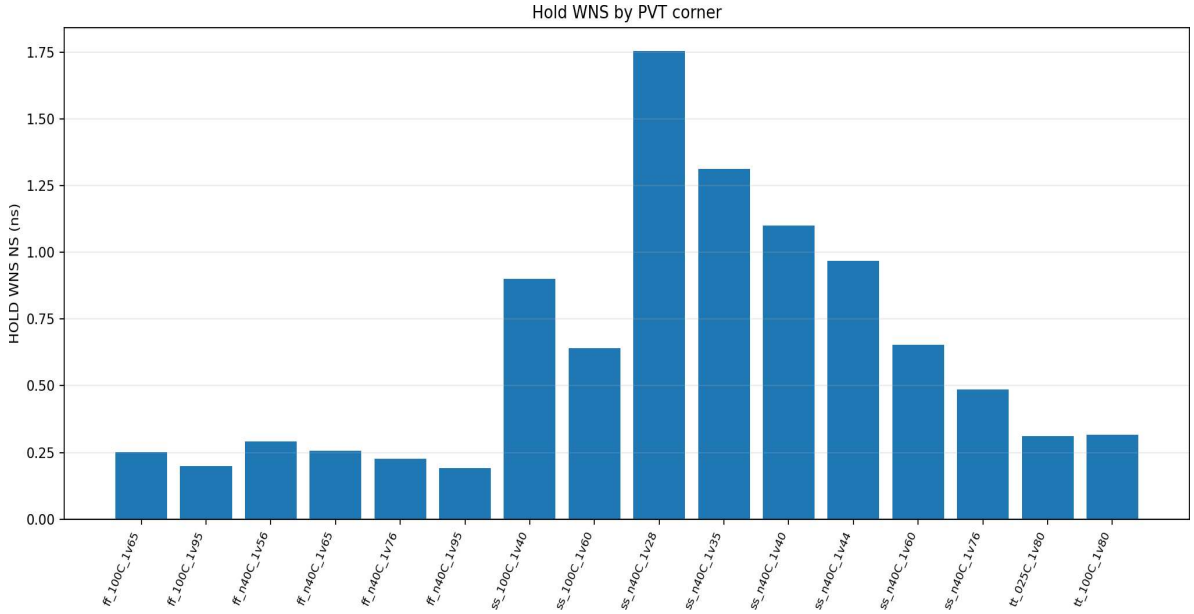
Setup WNS by corner



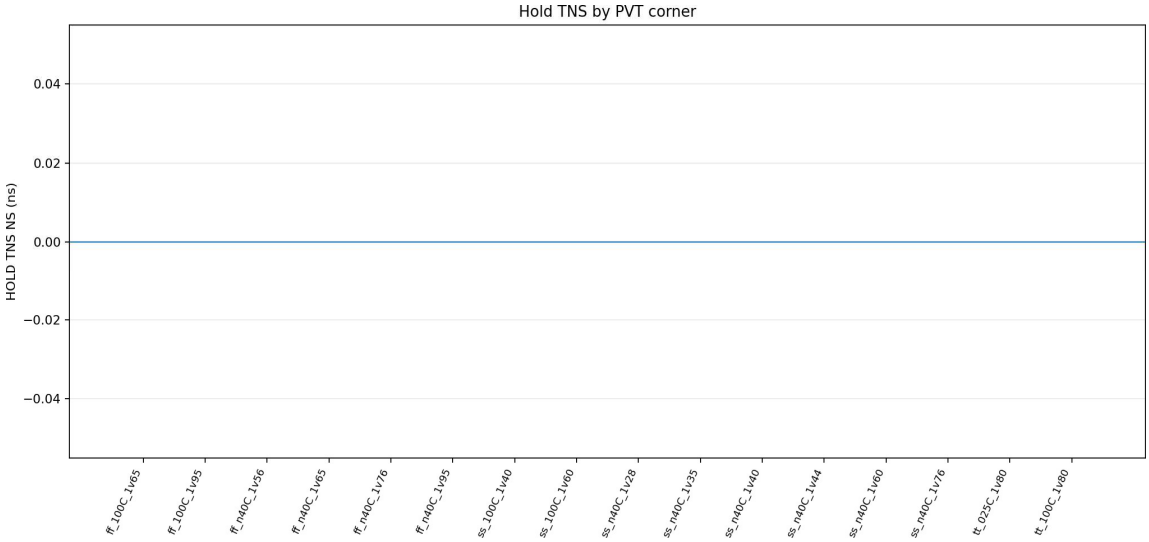
Setup TNS by corner



Hold WNS by corner



Hold TNS by corner



Interpretation and limitations

Six corners show negative setup WNS, with the most severe result at ss_n40C_1v28. All listed hold WNS values are positive and all hold TNS values are zero. The run status field is not a timing pass/fail flag.